

WEEKLY REPORT – DIGITAL SYSTEM DESIGN USING VERILOG & FPGA

1. General Information

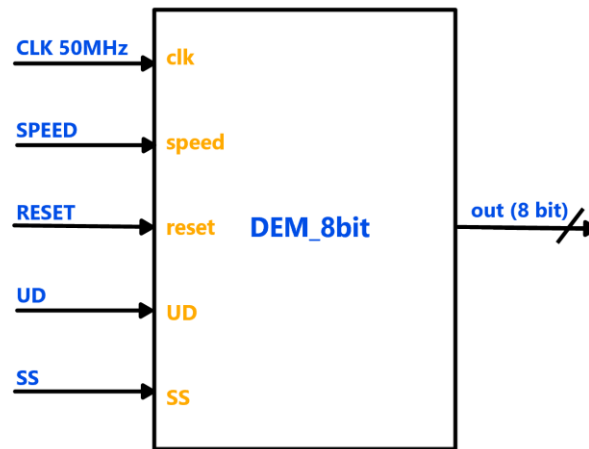
Project Title: 8-bit up and down counter RTL design and simulation

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2. Block Diagram



3. State Transition Diagram

The **DEM_8bit** module is an 8-bit up and down counter controlled by the **reset**, **SS**, **UD**, and **speed** signals. The input clock frequency is **50 MHz**

reset	SS	UD	speed	Function
1	X	X	X	Reset counter: OUT = 0
0	0	X	X	Stop counting; OUT holds its value
0	1	1	0	Count up
0	1	0	0	Count down
0	1	1	1	Count up
0	1	0	1	Count down

Control signal description:

- **RESET = 1:** Reset the 8-bit output to 0.
- **SS = 0:** stop counting and keep the current output value.
- **SS = 1:** enable counting.
- **UD = 1:** count upward (out = out + 1).
- **UD = 0:** count downward (out = out - 1).
- **speed = 0:** select the higher counting speed.
- **speed = 1:** select the lower counting speed.

4. Test Cases Overview

TC ID	Purpose	Input	Expected Output	Result (Pass/Fail)
TC1	Up counting at high speed (5MHz)	RESET = 0, SS = 1, UD = 1, speed = 0	OUT increases by 1 continuously	Pass
TC2	Stop counting	SS = 0	OUT remains unchanged	Pass
TC3	Down counting	SS = 1, UD = 0, speed = 0	OUT decreases by 1 continuously	Pass
TC4	Up counting at low speed (1MHz)	SS = 1, UD = 1, speed = 1	OUT increases by 1 at the lower counting speed	Pass

5. Testcase Details

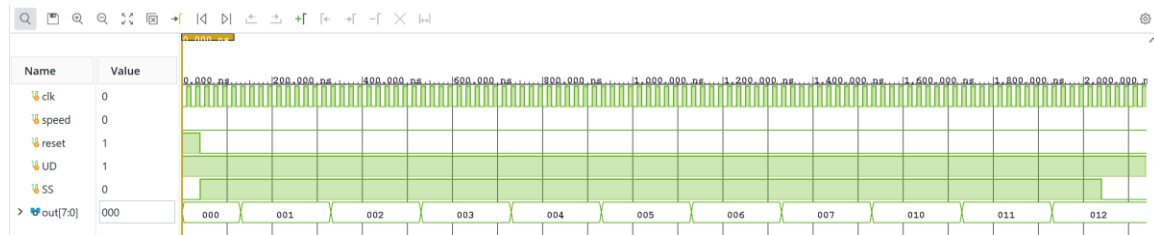
Testcase 1: Up counting

Purpose: to verify the up-counting function when speed = 0.

Input/Stimulus: reset = 0, SS = 1, UD = 1, speed = 0.

Expected Output: out increases sequentially by 1 (0 → 1 → 2 → 3 → ... → 12) at a counting frequency of 5 MHz

Waveform Simulation:



Analysis: when SS = 1, the counter is enabled. Since UD = 1, the counter operates in the up-counting mode. With speed = 0, the counting frequency is 5 MHz, so OUT increases by 1 at each counting edge.

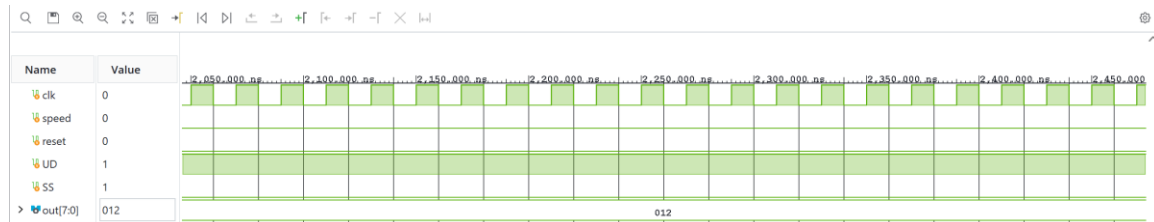
Testcase 2: stop counting

Purpose: to verify the stop/hold function of the counter.

Input/Stimulus: SS = 0.

Expected Output: out remains unchanged while SS = 0.

Waveform Simulation:



Analysis: when SS = 0, counting is disabled. The output out holds its previous value even though the input clock continues running.

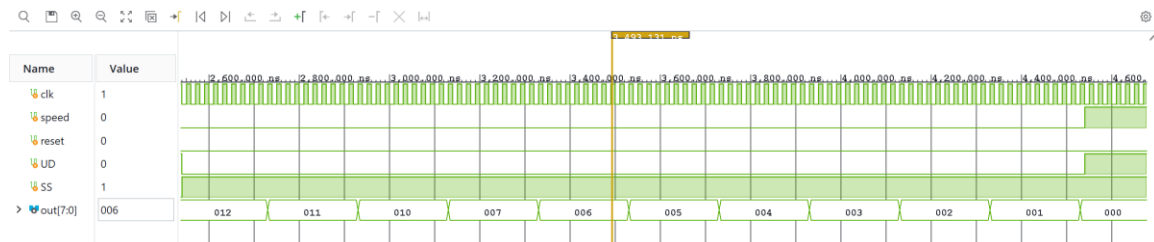
Testcase 3: down counting

Purpose: to verify the down-counting function.

Input/Stimulus: reset = 0, SS = 1, UD = 0, speed = 0.

Expected Output: out decreases sequentially by 1 (12 $\rightarrow$ 11 $\rightarrow$... $\rightarrow$ 0) at a counting frequency of 5 MHz

Waveform Simulation:



Analysis: when SS = 1, the counter is enabled. Since UD = 0, the counter operates in the down-counting mode. Therefore, out decreases by 1 at each counting edge.

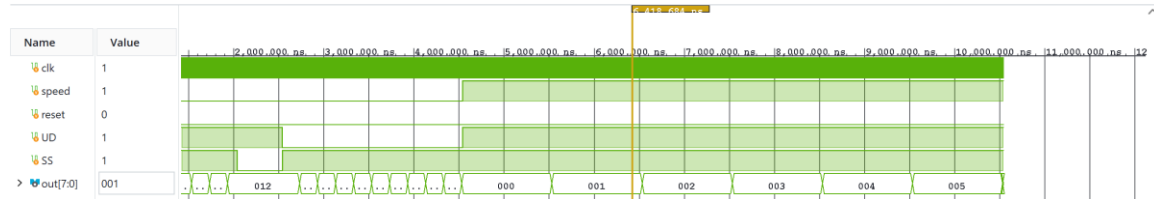
Testcase 4: speed selection

Purpose: to verify the speed selection function.

Input/Stimulus: SS = 1, UD = 1, speed = 1.

Expected Output: out continues to increase by 1, but at a lower counting frequency of 1 MHz.

Waveform Simulation:



Analysis: when speed = 1, the clock divider selects the lower counting frequency. The counting frequency changes from 5 MHz to 1 MHz, resulting in a longer time interval between consecutive changes of out.

7. Summary

The DEM_8bit module was successfully designed and simulated as an 8-bit up/down counter. The design supports reset, up-counting, down-counting, stop/hold operation, and selectable counting speeds of 5 MHz and 1 MHz through the speed control signal. The simulation test cases verify the main functions of the design, including up counting, stopping, down counting, and speed selection. The obtained waveforms demonstrate that the output out responds correctly to the control signals and follows the expected counting behavior.